

Cornell Bowers C-IS

College of Computing and Information Science

Deep Learning

Week 7: The Variational Auto-Encoder (VAE)
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Aggarwal, Oliver Richardson

Logistics

- **HW3** is due today (03/12)
- **A3** due 03/21
- **HW4** will be released today (03/12) - due next Thursday (03/19)
- **Paper Quiz 3** is released today (03/12), and due today (03/12)
- **Midterm** soon (03/24)!, review session 03/21 at 10am in Uris G01

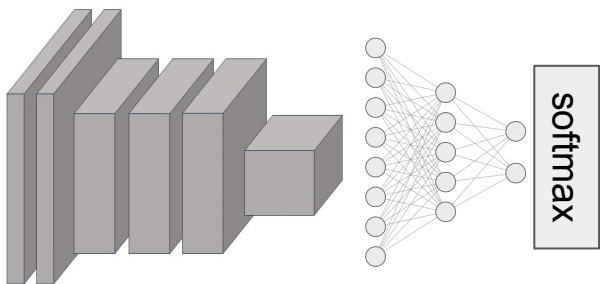
Discriminative Models

typically supervised

Goal: model $p(Y|X)$

from samples of $p(X,Y)$

(* so that we can predict
most likely labels)



Generative Models

unsupervised

Goal: model $p(X)$

from samples of $p(X)$

(* so that we can
generate artificial/new data)

Examples:

- GANs + variants
- Normalizing Flow Models
- Variational Autoencoders
 - Diffusion Models

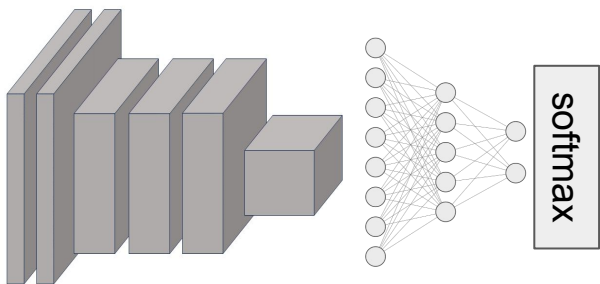
Discriminative Models

typically supervised

Goal: model $p(Y|X)$

from samples of $p(X,Y)$

(* so that we can predict
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Generative Models

(Conditional generation)

Goal: model $p(X|Y)$

from samples of $p(X,Y)$

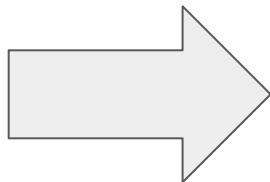
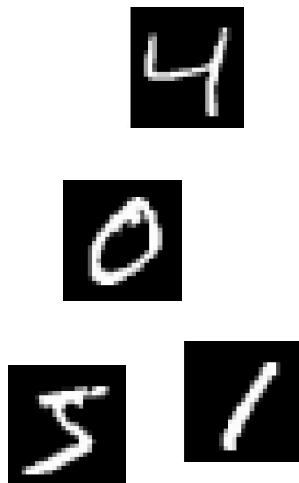
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Examples:

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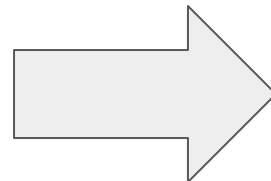
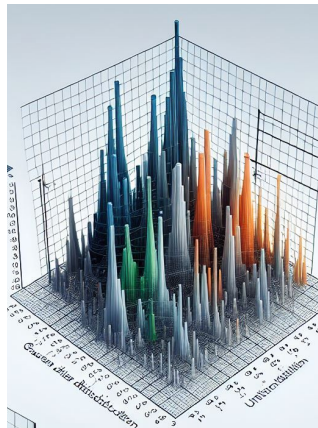
Big Picture

Data sampled from
true (but elusive) $P(X)$



Learn approximate
data distribution

$$Q(X) \approx P(X)$$

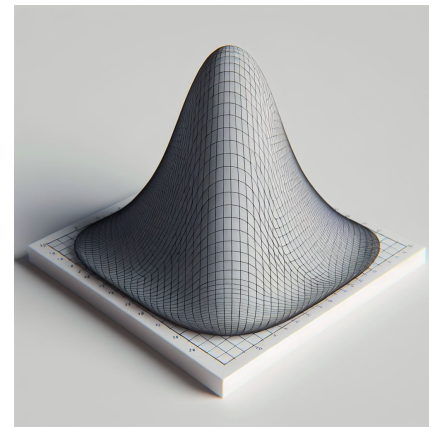
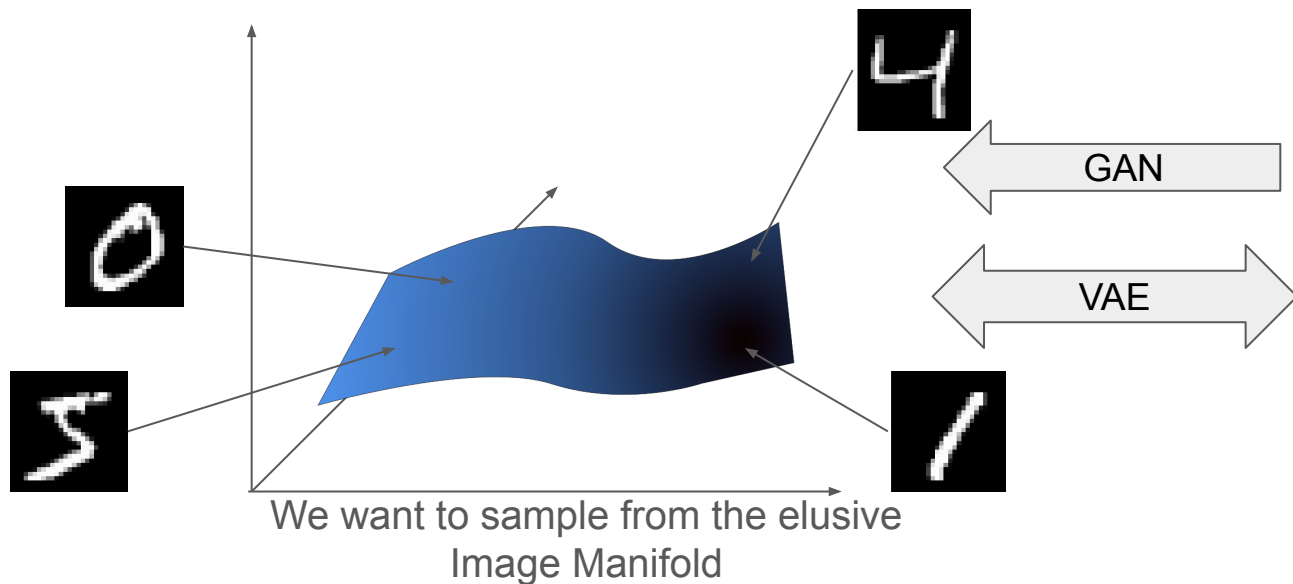


New (fake) data drawn
from $Q(x)$



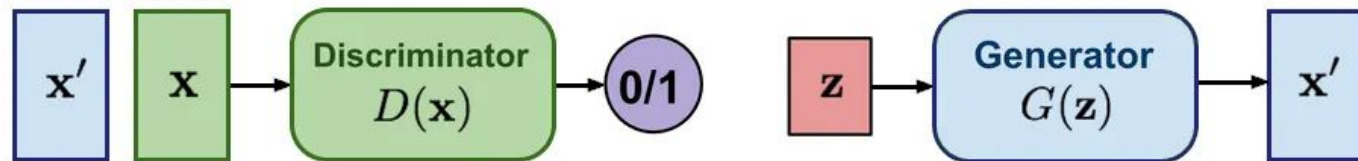
Data Manifold

- Data distribution $\mathbf{P}(\mathbf{X})$ defines a manifold of valid images
- Problem: data manifold takes up **tiny** volume of ambient space
- Naive random samples (e.g. within $[0,1]^d$) are always **off manifold**
- Solution: Sample from a Gaussian, then learn mapping to and from manifold

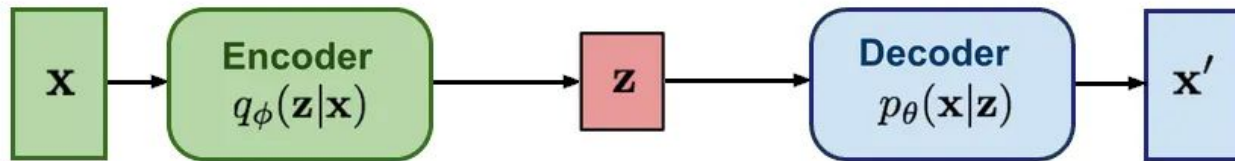


We **can** sample from a Gaussian Distribution

GAN: Adversarial training



VAE: maximize variational lower bound

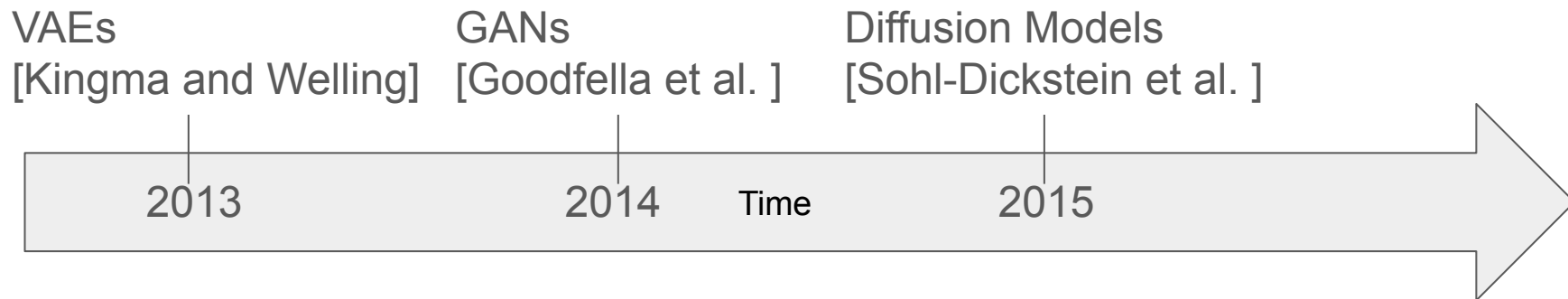


Diffusion models:
Gradually add Gaussian noise and then reverse



Timeline

- VAEs preceded GANs.
- In fact, GANs were motivated to fix some of the problems of VAEs.
- VAEs are important to understand **Diffusion Models**



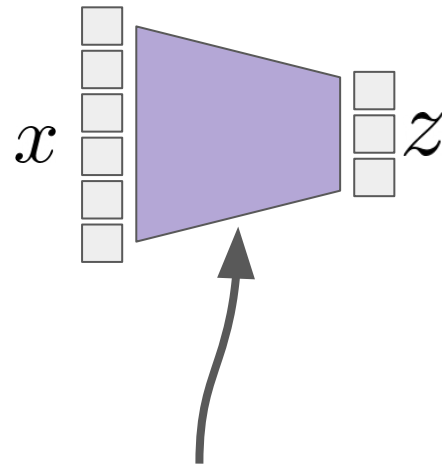
Dimensionality Reduction

Data is typically from a **low dimensional** distribution embedded in a much **higher dimensional** ambient space.

Want to map images $x \in \mathbb{R}^D$
to low-dimensional $z \in \mathbb{R}^d$

Often for the purposes of

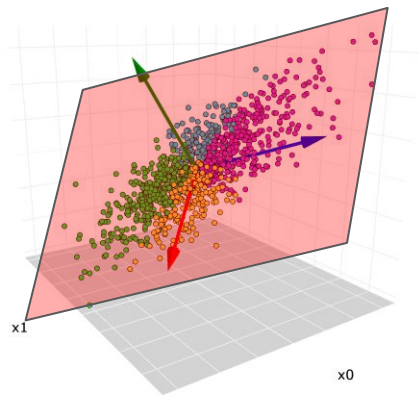
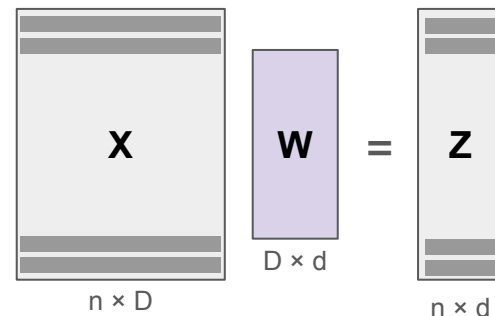
- visualization
- extracting important features
(for downstream tasks)
- representing meaningful
relationships between samples
- In this lecture sampling!



Question: what properties should this mapping have?

Principal Component Analysis (PCA)

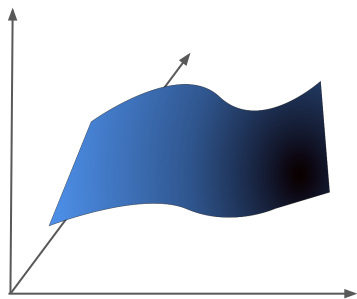
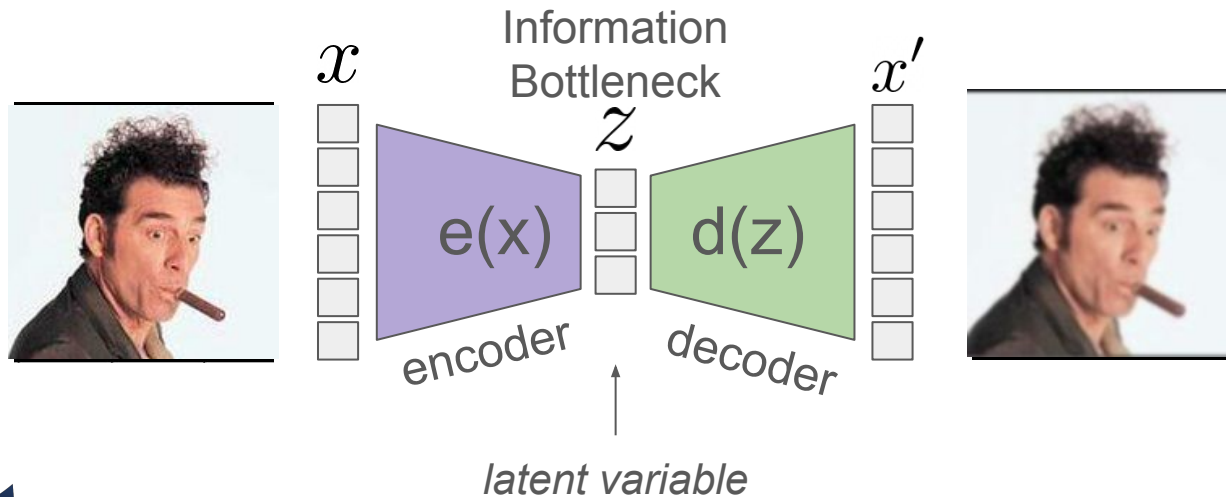
- **assumption: data manifold is a subspace**
- $\mathbf{z} = \mathbf{W}(\mathbf{x} - \mu)$ (a linear transformation)
- $\mathbf{x} \approx \mathbf{W}^\top \mathbf{z} + \mu$ (reconstruction)
- capture as much variance as possible



Can be computed directly with linear algebra:
take leading eigenvectors of (centered) scatter
matrix $\mathbf{X}\mathbf{X}'$!

Autoencoders [Kramer, 1991]

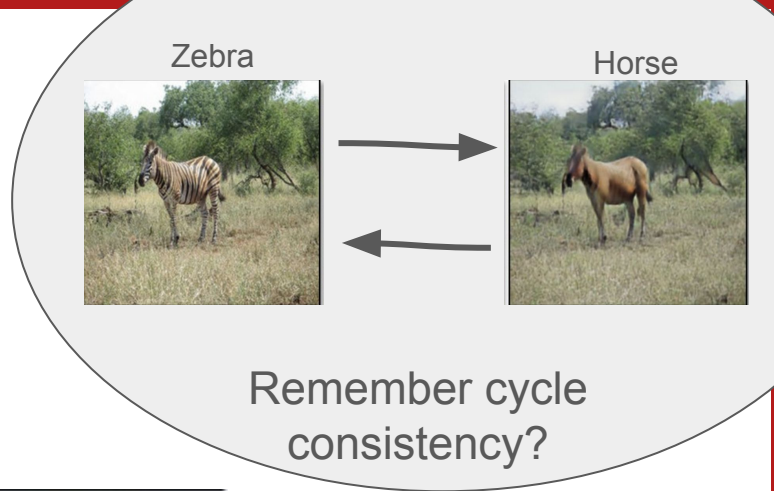
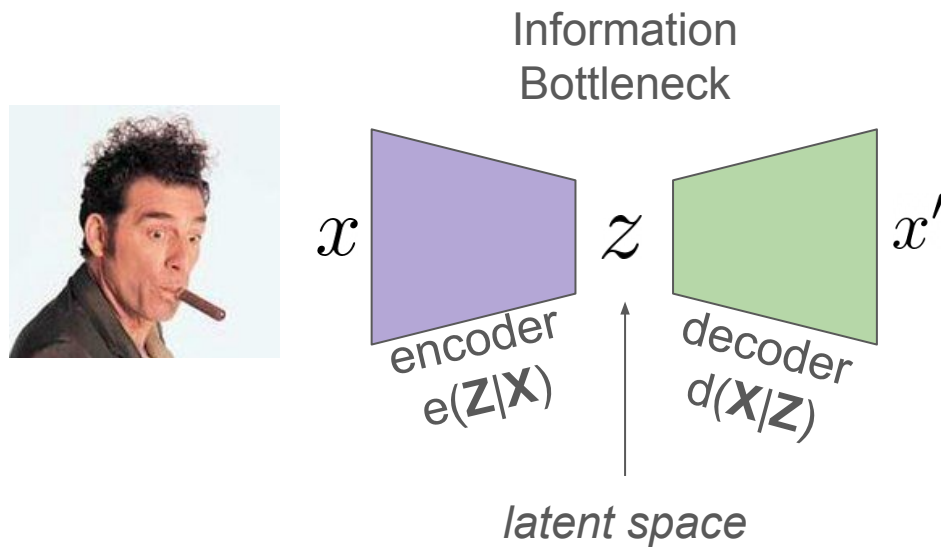
Non-linear dimensionality reduction.



Question: What loss function should we use to learn $e()$ and $d()$?
What happens if $e(x)$ and $d(z)$ are both linear functions?

Autoencoders [Kramer, 1991]

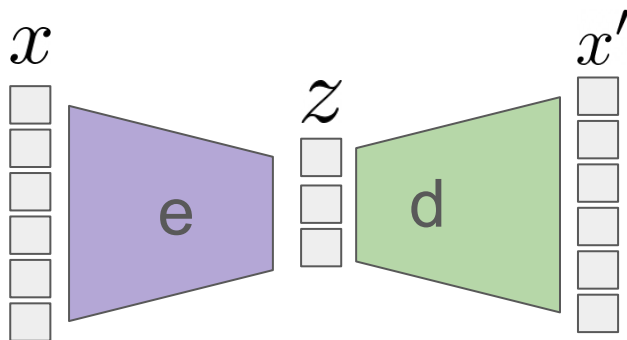
Typical loss: Squared loss, or absolute loss



$$\min_{\phi, \theta} \sum_{\mathbf{x} \in \mathcal{D}} (\mathbf{x} - \mathbf{x}')^2$$

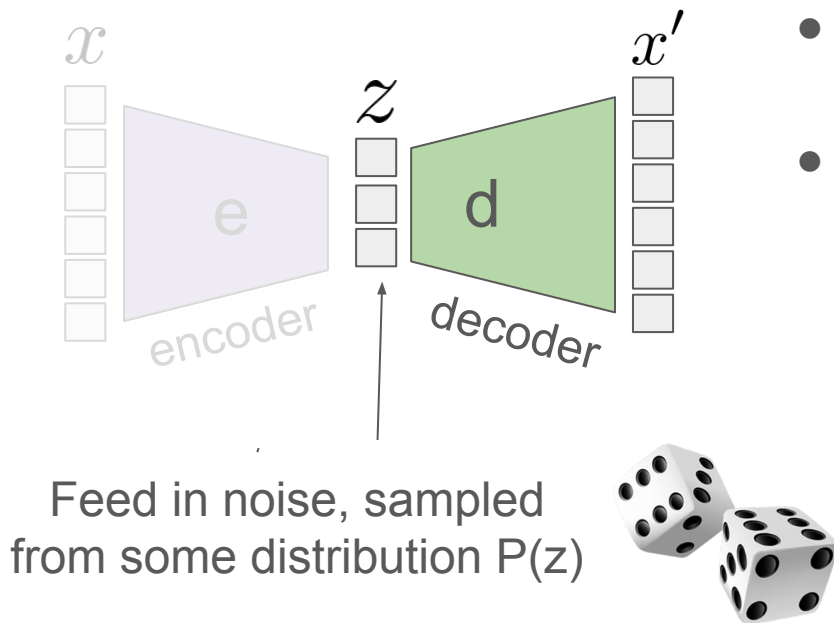
$$\mathbf{x}' = d_{\theta}(e_{\phi}(\mathbf{x}))$$

Idea: Sampling from a trained Autoencoder



- GANs train the decoder with a discriminator
- VAEs ensure quality with
 - Reconstruction loss
 - KL regularization (in a few slides)

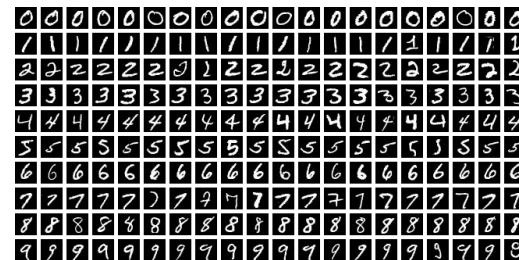
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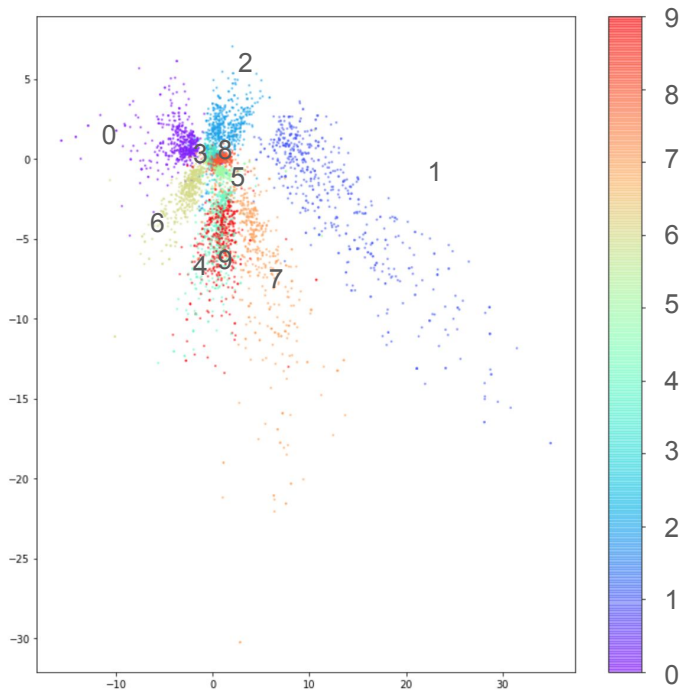
- GANs train the decoder with a discriminator
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Crucial insight:

We can amend latent space so that it is easy to sample from it.



Autoencoder trained on MNIST: latent space



Naive representation (without any special effort), not favorable:

- lots of empty space
- no symmetries between digit representations
- **Not easy to sample in latent space**

Figure 3-8. Plot of the latent space, colored by digit

Naive sampling in latent space does not work



reconstructed sample

$$x' = d(e(x))$$



new image?

$$x' = d(\text{noise})$$

Some Fundamentals of probability and information

Building Blocks:

- Conditional and marginal probabilities
- Surprisal / Negative Log Likelihood
- Relative Entropy / KL Divergence

Conditional and Marginal Probabilities

$$p(X, Y) = p(Y|X)p(X)$$

$$p(X) = \int p(X, y) \, dy$$

Equation (2): Quick 60s Stats Puzzle

Prove that for any random variable X, Z .

$$p(\mathbf{x}) = \frac{p(\mathbf{x}, \mathbf{z})}{p(\mathbf{z}|\mathbf{x})}$$

Hint: Later this rule will come in handy. Remember it as “Equation (2)”.

Equivalences

$$\begin{aligned} \min_{\theta} \log \left(\frac{1}{p_{\theta}(x)} \right) &= \min_{\theta} -\log(p_{\theta}(x)) \\ &= \max_{\theta} \log(p_{\theta}(x)) \end{aligned}$$

surprisal

Negative log-likelihood

log-likelihood

KL Divergence (a.k.a. relative entropy)

$$D(p \parallel q) := \mathbb{E}_{x \sim p} \left[\log \frac{p(x)}{q(x)} \right]$$

reality (e.g., dataset) model

$$\mathbb{E}_{x \sim p} [\log p(x)] - \log q(x)$$

likelihood in real world; does not depend on model q Likelihood under model

- non-negative $D(p \parallel q) \geq 0$
- zero means same $D(p \parallel q) = 0 \iff p = q$
- not symmetric
- has many other, uniquely nice properties...

KL Divergence

Wei-Chiu's Coin



Kilian's Coin



Question:

What is higher $KL(\text{Wei-Chiu} \parallel \text{Kilian})$ or $KL(\text{Kilian} \parallel \text{Wei-Chiu})$?

[\[link to visualization \]](#)

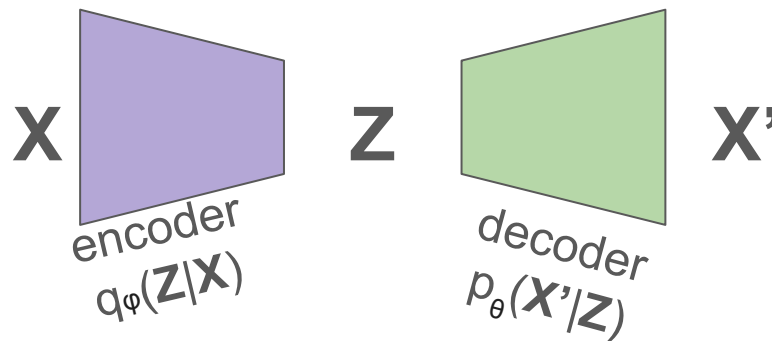
Building Blocks:

- ✓ Conditional and marginal probabilities
- ✓ Surprisal / Negative Log Likelihood
- ✓ Relative Entropy / KL Divergence

VAEs, Step 1: Make AutoEncoder probabilistic

Reconstruction Loss, using surprisal

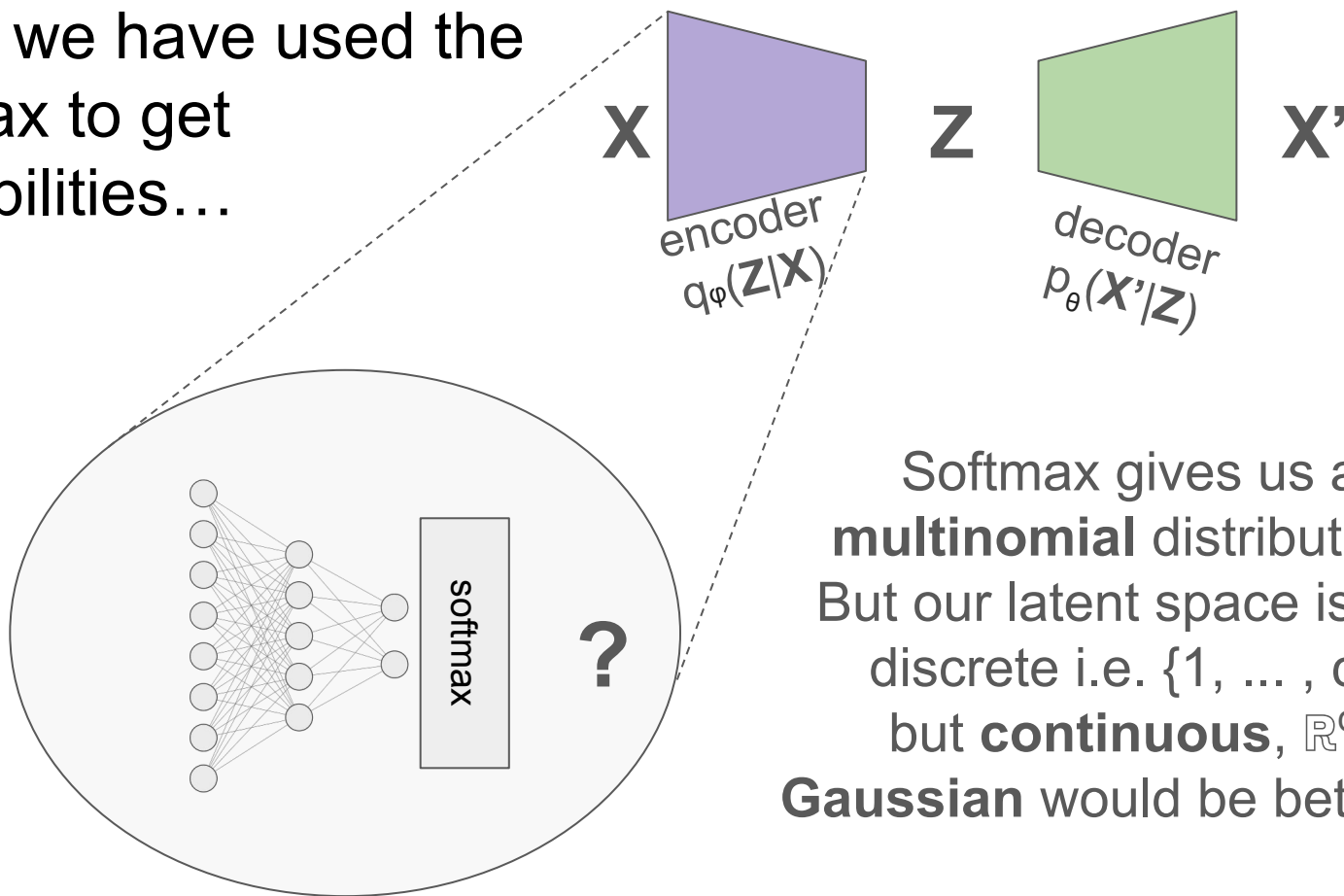
Back to our AutoEncoder,
but this time we make
everything **probabilistic**!



$$\max_{\phi, \theta} \mathbb{E}_{\mathbf{z} \sim q_{\phi}(\mathbf{z}|\mathbf{x})} [\log p_{\theta}(\mathbf{x} | \mathbf{z})]$$

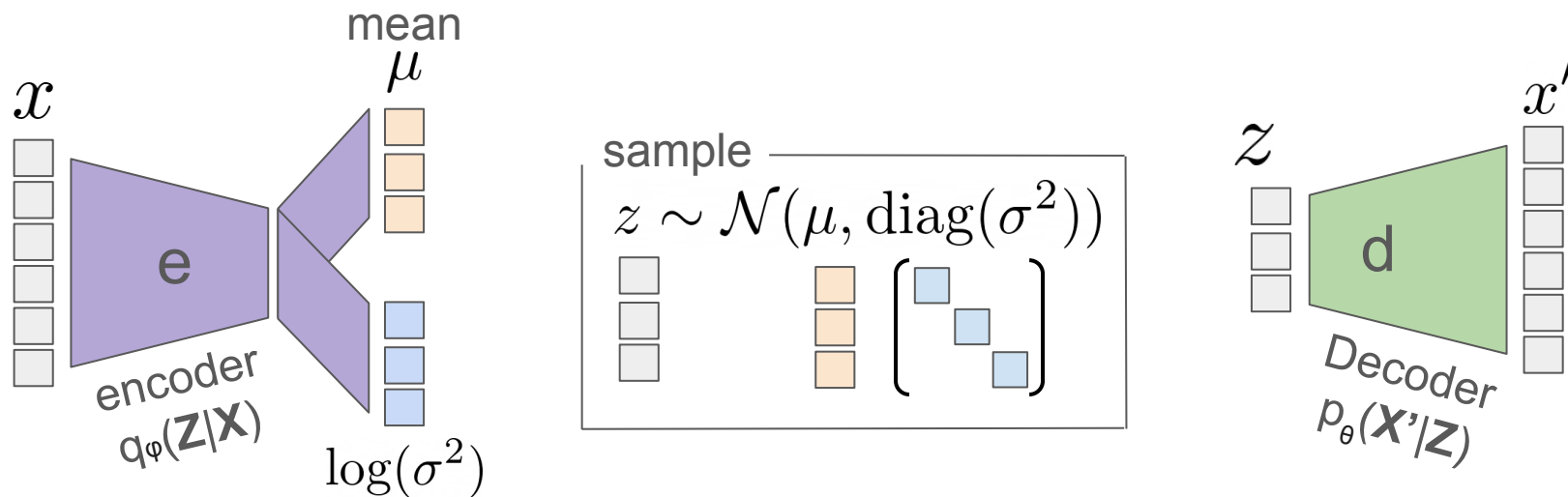
How likely would it be to encode $\mathbf{x}$,
decode the result, and recover $\mathbf{x}$?

So far we have used the softmax to get probabilities...



Softmax gives us a **multinomial** distribution. But our latent space is not discrete i.e. $\{1, \dots, c\}$, but **continuous**, $\mathbb{R}^d$! **Gaussian** would be better ...

Probabilistic **Encoder** (Gaussian)

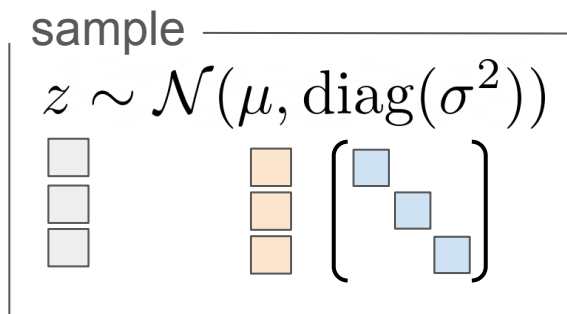


Problem: backpropagation through sampling process?

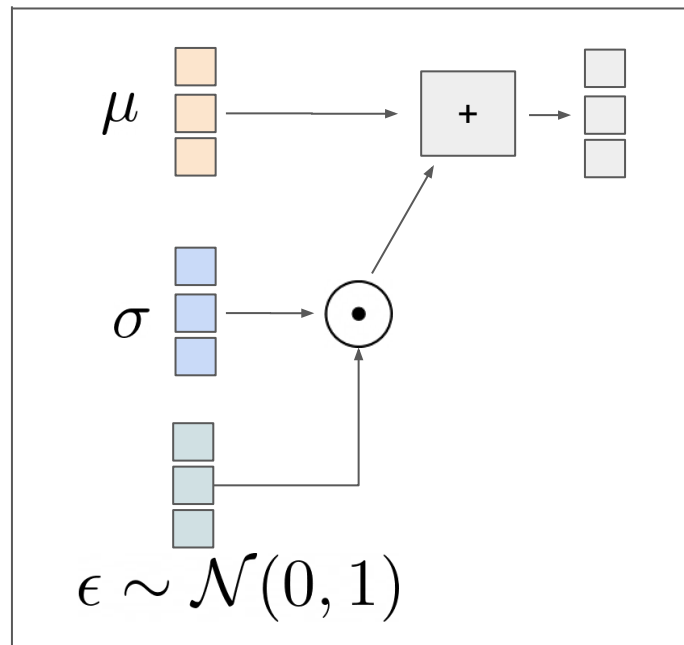
$$\max_{\phi, \theta} \mathbb{E}_{z \sim q_\phi(\mathbf{z}|\mathbf{x})} [\log(p_\theta(\mathbf{x}|\mathbf{z}))]$$

The Reparameterization Trick

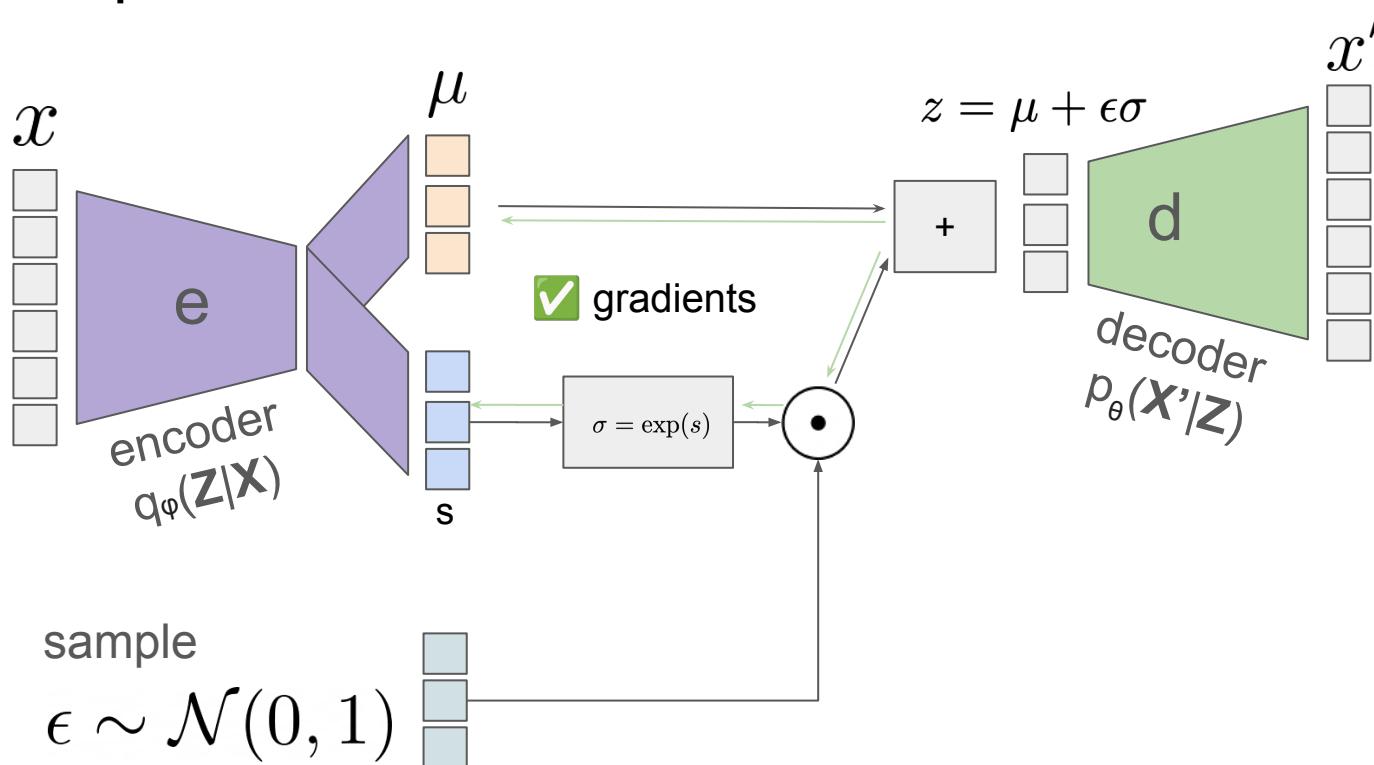
$$\mathcal{N}(\mu, \text{diag}(\sigma^2)) = \mu + \sigma \odot \mathcal{N}(0, I)$$



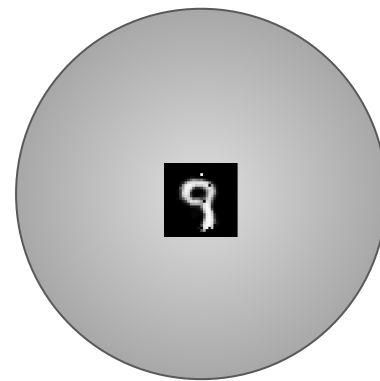
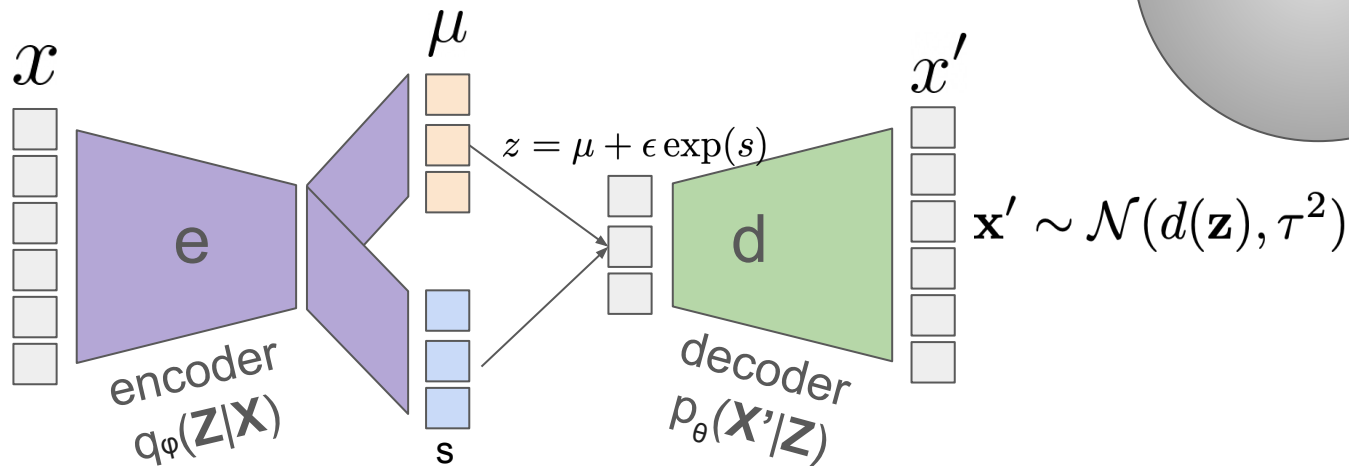
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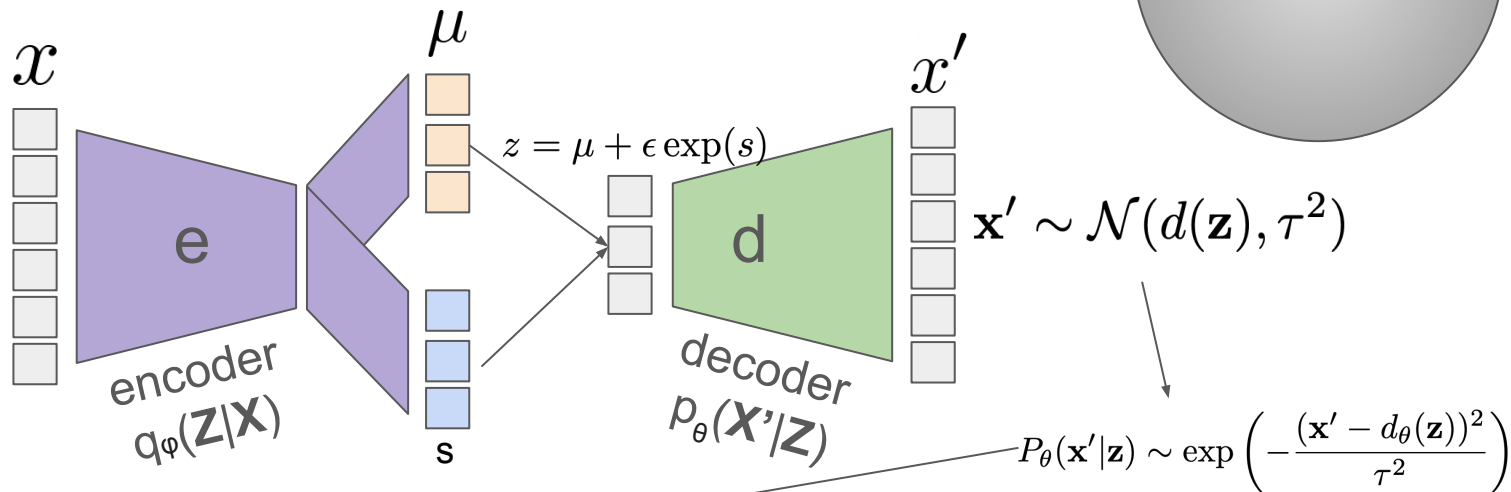
The Reparameterization Trick



Probabilistic **decoder** (Gaussian)



Probabilistic **decoder** (Gaussian)

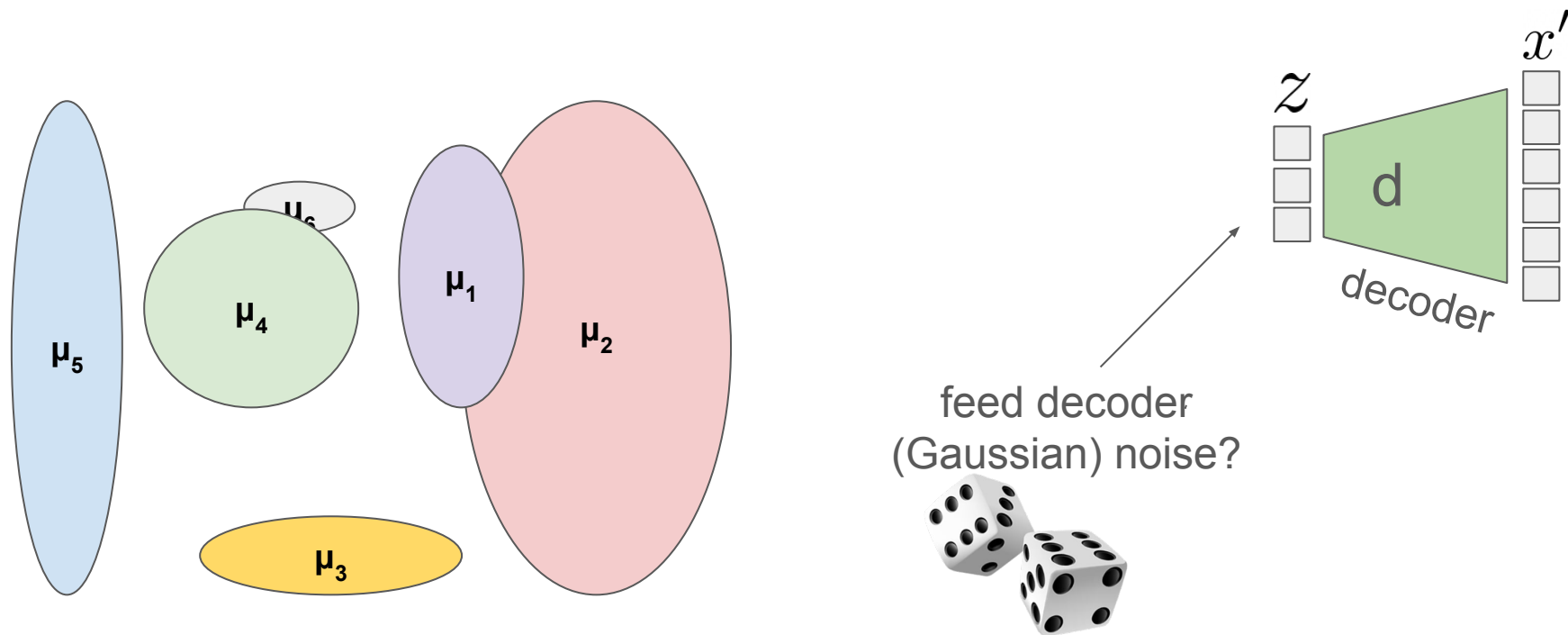


$$\max_{\phi, \theta} \mathbb{E}_{z \sim q_\phi(\mathbf{z}|\mathbf{x})} [\log(p_\theta(\mathbf{x}|\mathbf{z}))] = \min_{\phi, \theta} \mathbb{E}_{z \sim q_\phi(\mathbf{z}|\mathbf{x})} [(\mathbf{x} - d_\theta(\mathbf{z}))^2]$$

Plugging the output distribution into the reconstruction loss, results in the squared loss.

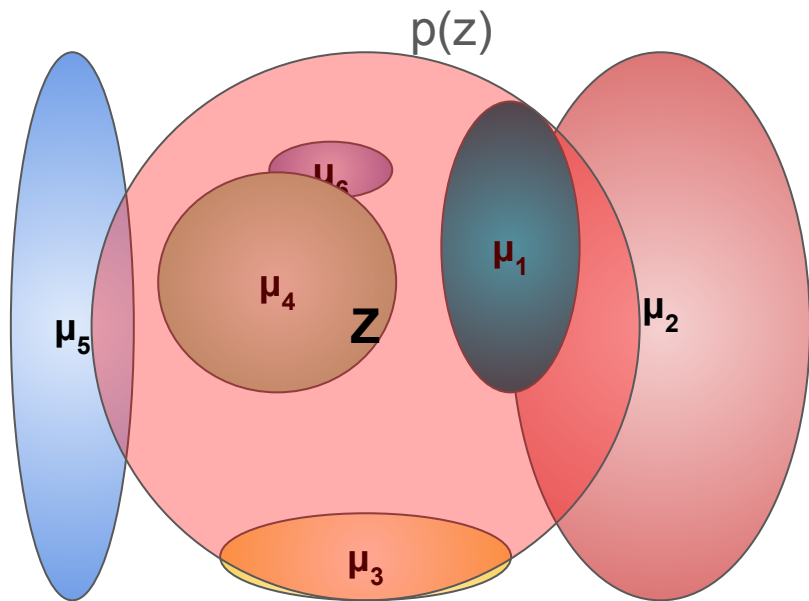
Step 2: How do we sample in latent space?

How can we sample, if each sample has its own latent distribution?



Step 2: How do we sample in latent space?

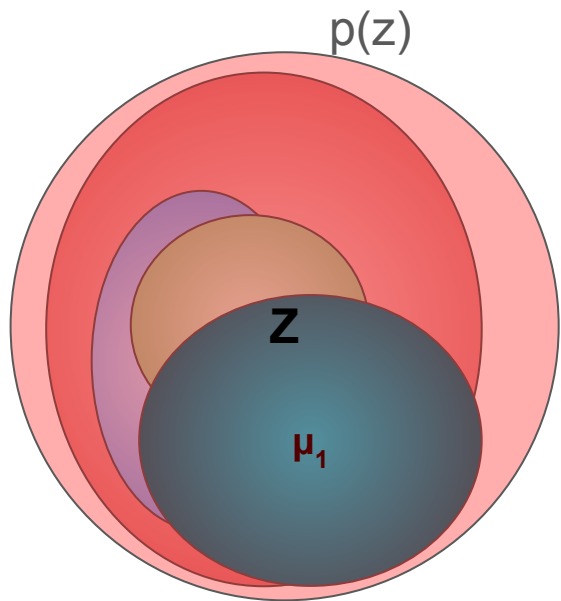
Solution: Regularize all distributions to be close to the standard normal $\mathcal{N}(\mathbf{0}; \mathbf{I})$.



$$\text{maximize} \quad \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}))}_{\text{prior matching term}}$$

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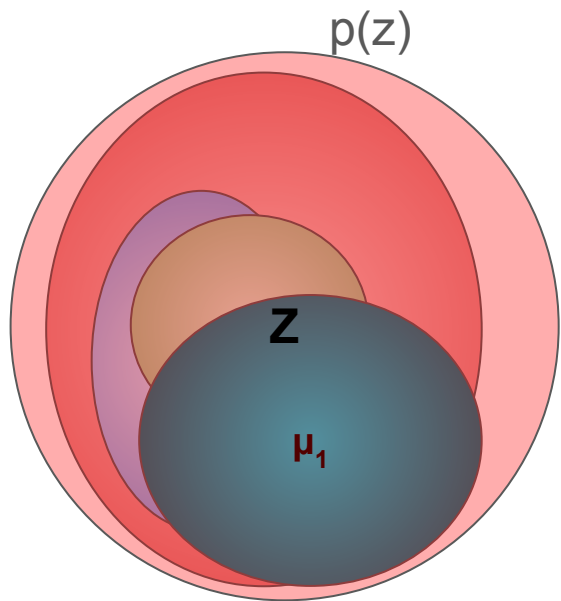
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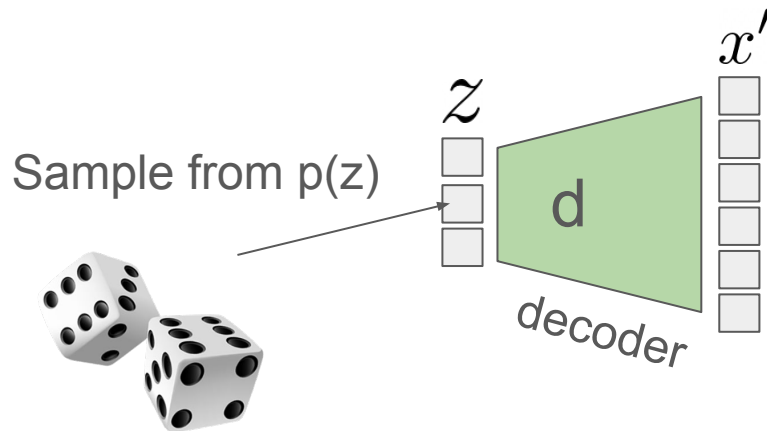
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$$\text{maximize} \quad \underbrace{\mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q_\phi(z|x) \parallel p(z))}_{\text{prior matching term}}$$



Evidence Lower Bound (ELBO)

$$\log p(\mathbf{x}) = \log p(\mathbf{x}) \int q_{\phi}(\mathbf{z}|\mathbf{x}) d\mathbf{z}$$

(Multiply by $1 = \int q_{\phi}(\mathbf{z}|\mathbf{x}) d\mathbf{z}$)

$$= \int q_{\phi}(\mathbf{z}|\mathbf{x}) (\log p(\mathbf{x})) d\mathbf{z}$$

(Bring evidence into integral)

$$= \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} [\log p(\mathbf{x})]$$

(Definition of Expectation)

$$= \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{p(\mathbf{z}|\mathbf{x})} \right]$$

(Apply Equation 2)

$$= \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z}) q_{\phi}(\mathbf{z}|\mathbf{x})}{p(\mathbf{z}|\mathbf{x}) q_{\phi}(\mathbf{z}|\mathbf{x})} \right]$$

(Multiply by $1 = \frac{q_{\phi}(\mathbf{z}|\mathbf{x})}{q_{\phi}(\mathbf{z}|\mathbf{x})}$)

$$= \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \right] + \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[\log \frac{q_{\phi}(\mathbf{z}|\mathbf{x})}{p(\mathbf{z}|\mathbf{x})} \right]$$

(Split the Expectation)

$$= \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \right] + D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}|\mathbf{x}))$$

(Definition of **KL Divergence**)

$$\geq \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \right]$$

(KL Divergence always ≥ 0)

Evidence Lower Bound (ELBO)

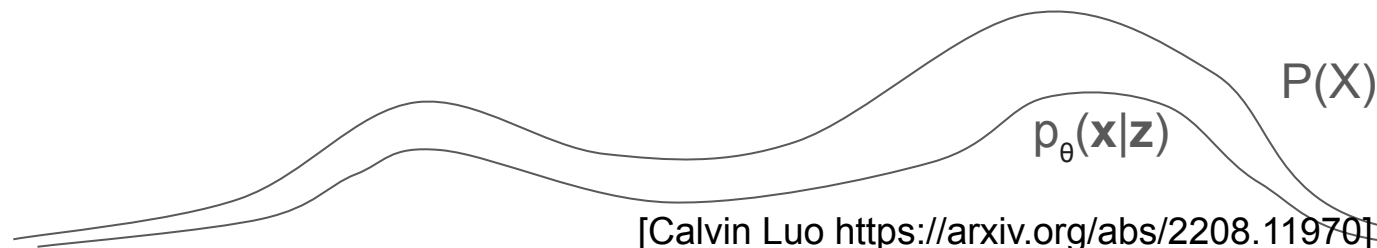
$$\mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{x}, \mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \right] = \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[\log \frac{p_{\theta}(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \right] \quad (\text{Chain Rule of Probability})$$

$$= \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} [\log p_{\theta}(\mathbf{x}|\mathbf{z})] + \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} \left[\log \frac{p(\mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \right] \quad (\text{Split the Expectation})$$

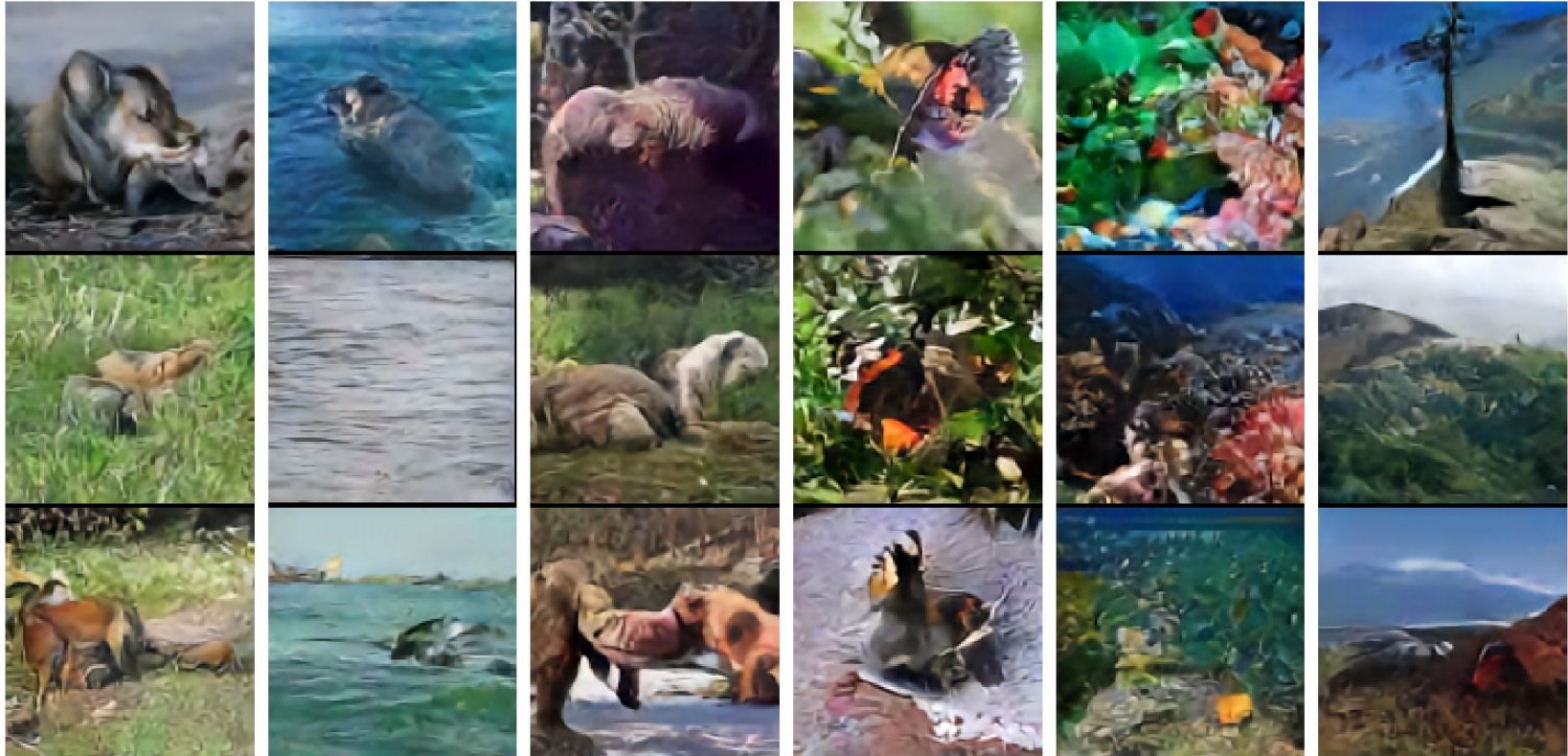
$$= \underbrace{\mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} [\log p_{\theta}(\mathbf{x}|\mathbf{z})]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}))}_{\text{prior matching term}} \quad (\text{Definition of KL Divergence})$$

(We are **maximizing** this lower bound.)

If we maximize $p_{\theta}(\mathbf{x}|\mathbf{z})$ and minimize the D_{KL} we get close to $P(\mathbf{x})$.



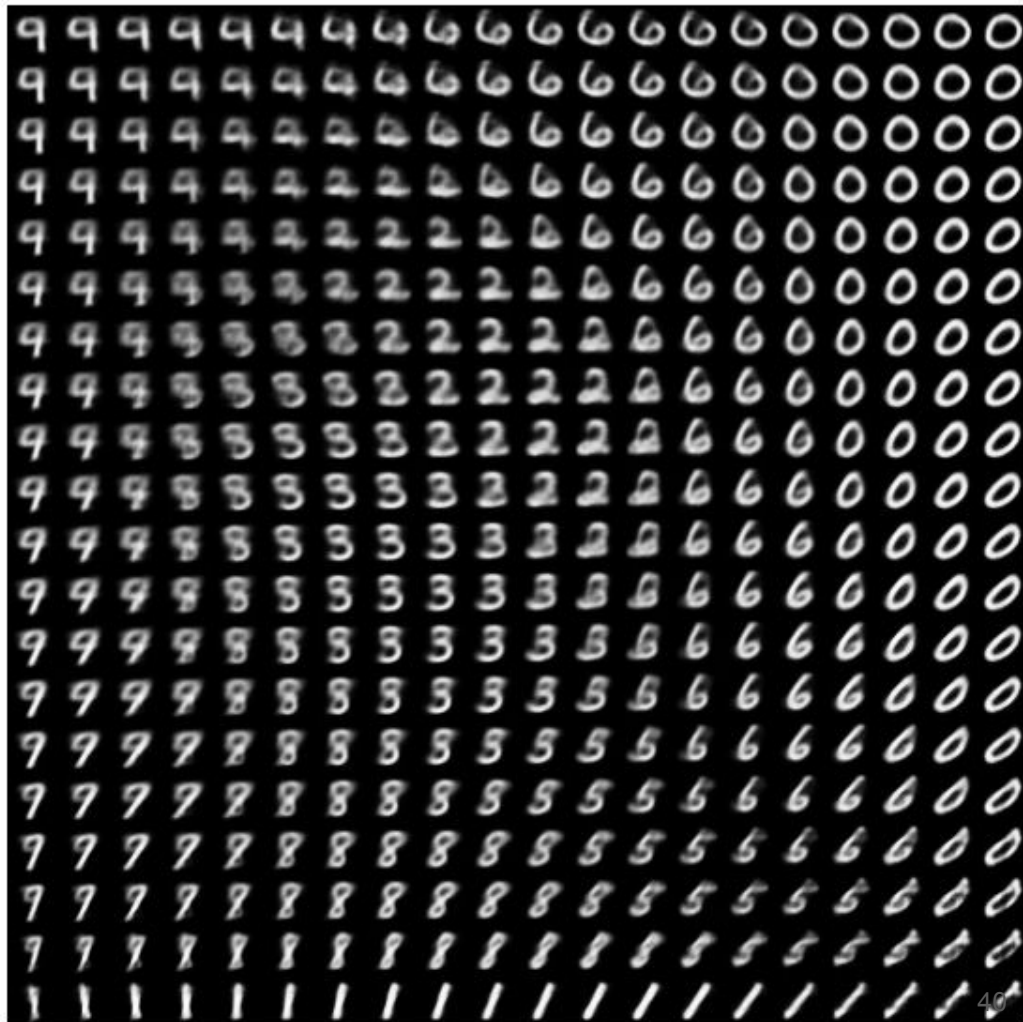
Examples of VAE generated images



a much nicer space...

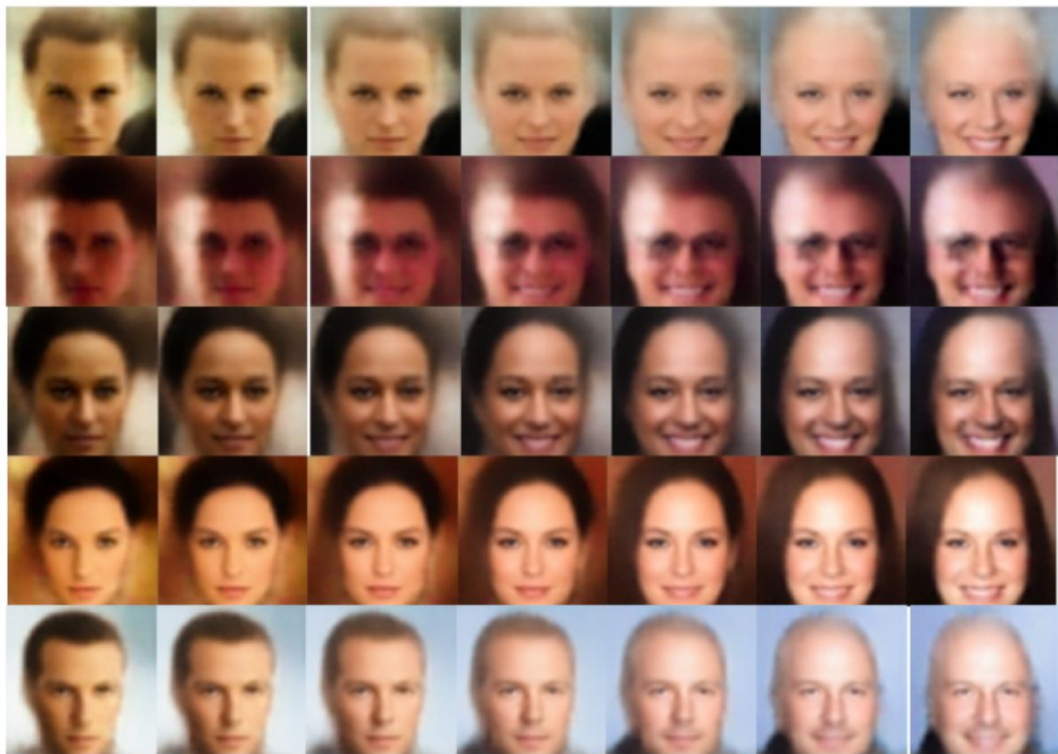
can smoothly interpolate digits in
a meaningful, digit-y kind of way

[DEMO](#)



a much nicer space

dimensions in latent space correspond to meaningful concepts, like sentiment and orientation



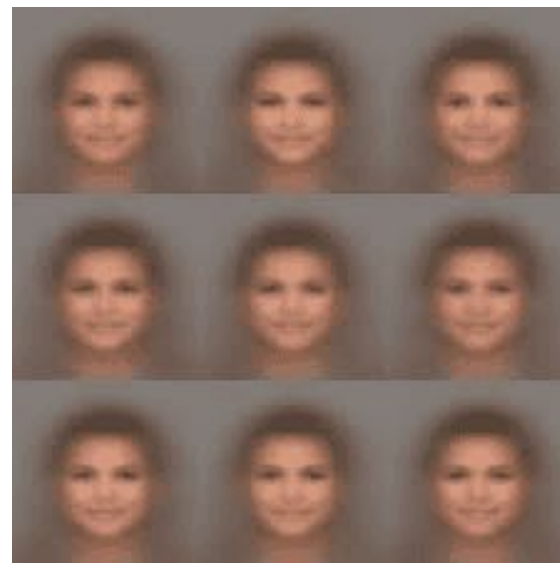
The Biggest Drawback of VAEs

- Out of the box, generated images can be blurry.

Question: Why? How do GANs fix this problem?

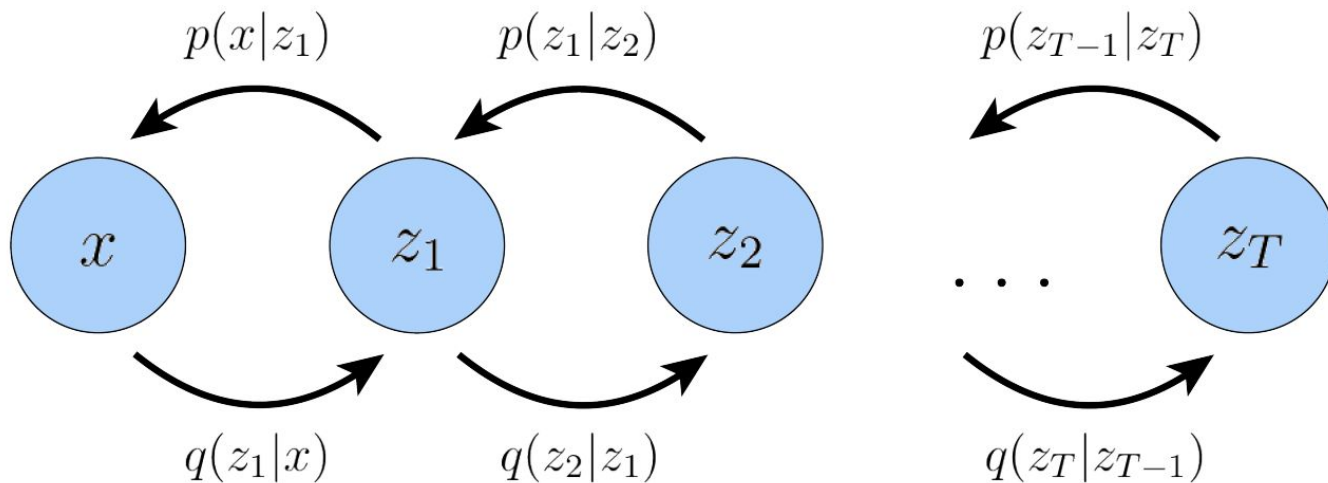


[VAE v. GAN](#)



Hierarchical VAEs

The generative process is modeled as a Markov chain, where each latent z_t is generated only from the previous latent z_{t+1}



Summary

- Generative Image models learn a mapping from the **Standard Normal Gaussian** to the **Image Manifold**
 - GANs learn this through a **discriminator**.
 - VAEs learn it through **variational autoencoders**
- AutoEncoders learn to **compress** and **reconstruct** data
- VAEs make these AutoEncoders **probabilistic**
 - Minimize the **reconstruction loss**
 - Latent space is sampled from Gaussian distributions
 - Sampling is made differentiable with the **Reparameterization Trick**
 - Deviations from the Prior (Standard Normal Gaussian) is penalized by **KL divergence**
- The ELBO is a **lower bound** of $P(X)$
 - Maximizing the ELBO, and minimizing the KL divergence makes $P(x|z)$ close to $P(x)$